

TOPIC 3

BPMN

Information Systems Analysis & Design

Learning Objectives

After this session, students will be able to:

- Explain the purpose of BPMN in business process modeling
- Identify BPMN core symbols: event, activity, gateway, flow, pool/lane
- Read and create a simple BPMN diagram
- Distinguish BPMN workflow models from DFD and architecture diagrams.

Why We Study BPMN in ISAD

Before designing a system, analysts must understand how work is performed.

BPMN helps analysts answer:

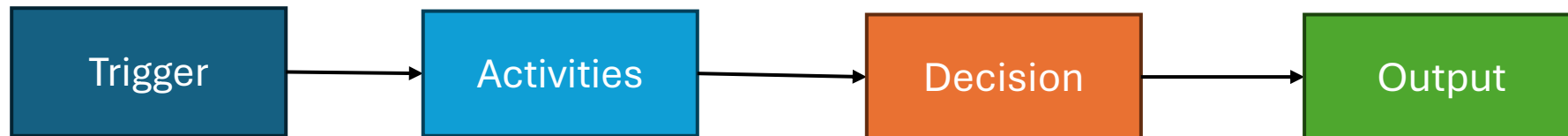
- What triggers the process?
- Who performs each step?
- What decisions are made?
- What happens when the process succeeds or fails?
- Where can the process be improved?

Business Process Concept

A **business process** is a structured set of activities performed to achieve a business goal.

Examples:

- Student submits an academic service request
- Customer places an online order
- Staff approves a leave request
- Warehouse prepares and ships an item



What is BPMN?

BPMN = Business Process Model and Notation

BPMN is a standardized graphical notation used to represent business processes as diagrams.

It shows:

- Sequence of work
- Participants involved
- Decisions and branches
- Messages between participants
- Process start and completion

Why Use BPMN?

Easier to understand than text descriptions

Helps **identify**:

- Bottlenecks
- Missing steps
- Inefficiencies

Creates **shared understanding** between:

- Business users
- Developers

BPMN vs DFD

BPMN	DFD
Models workflow	Models data movement
Shows sequence of activities	Shows how data enters, moves, and leaves the system
Uses gateways for decisions	Does not emphasize decision paths
Shows responsibility through lanes	Shows external entities, processes, data stores
Good for business process modeling	Good for system/data analysis

Key idea:

BPMN answers “How does the work happen?”

while DFD answers “How does data move through the system?”

BPMN Core Element Groups

BPMN has four main groups:

1. **Flow Objects**

Events, Activities, Gateways

2. **Connecting Objects**

Sequence Flow, Message Flow, Association

3. **Swimlanes**

Pool, Lane

4. **Artifacts**

Data Object, Annotation, Group

BPMN Notation Cheat Sheet

Symbol	Name	Meaning
	Start Event	Process begins
	Intermediate Event	Something happens during the process
	End Event	Process finishes
	Task	Work performed
	Exclusive Gateway	One path only
	Parallel Gateway	All paths happen
	Sequence Flow	Order of work
	Message Flow	Communication across pools
	Data Object	Data/document used or produced

Events

An **event** is something that happens in a process.

Types:

Event Type	Symbol	Meaning
Start Event	○	Begins the process
Intermediate Event	⊙	Occurs during the process
End Event	●	Finishes the process

In BPMN, an event is not an activity. It represents something that happens, such as receiving a request, reaching a deadline, or completing a process.



Activities / Tasks

An **activity** represents work performed by a person, role, department, or system.

Examples:

- Fill request form
- Check document completeness
- Approve request
- Send notification
- Update request status

Rule:

Use verb + object naming.

Good:

- “Check documents”
- “Approve request”

Not good:

- “Documents”
- “Approval”

Gateways

A **gateway** controls process branching or merging.

Common types:

- **XOR Gateway**: only one path is selected
- **AND Gateway**: all paths happen in parallel
- **OR Gateway**: one or more paths may happen

Example:

“Documents complete?”

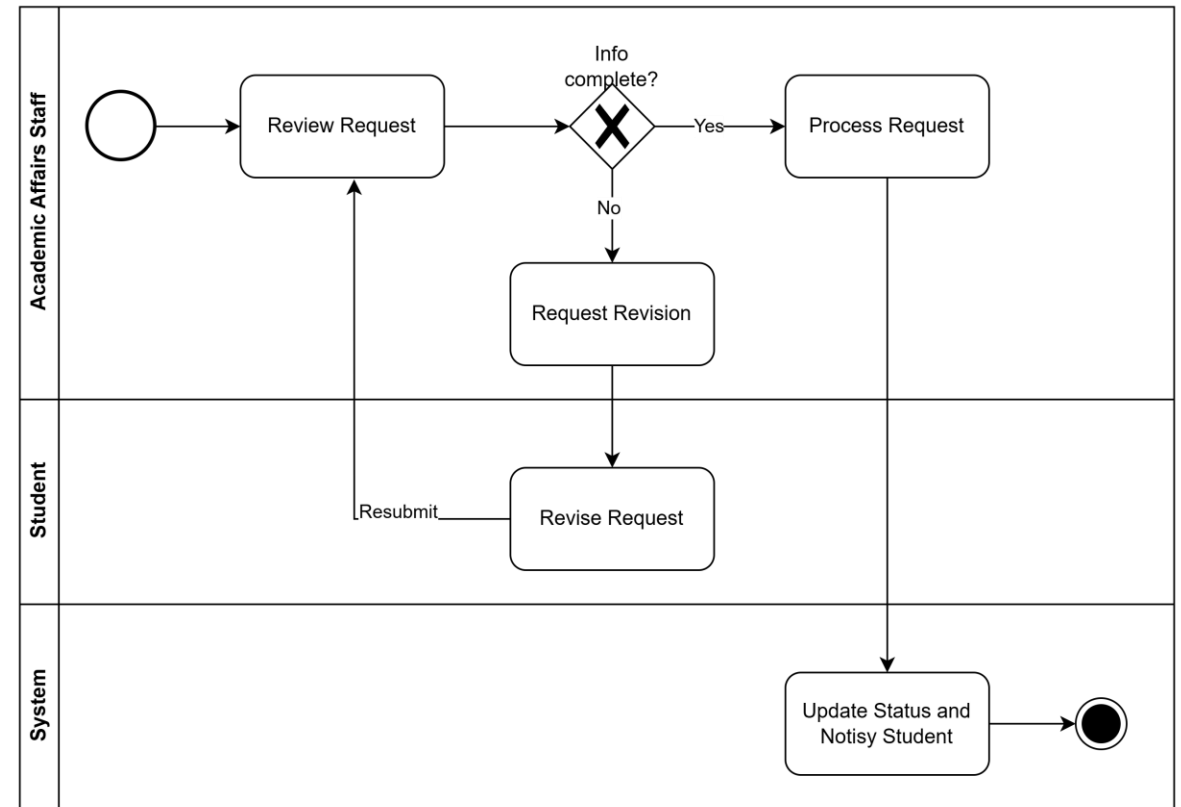
- Yes → Continue review
- No → Request correction

XOR Gateway Example

XOR Gateway is used when only one path can be selected based on a condition.

In this example, if the request information is complete, staff process the request. If not, staff ask the student to revise and resubmit

Gateway question: Is the request information complete?



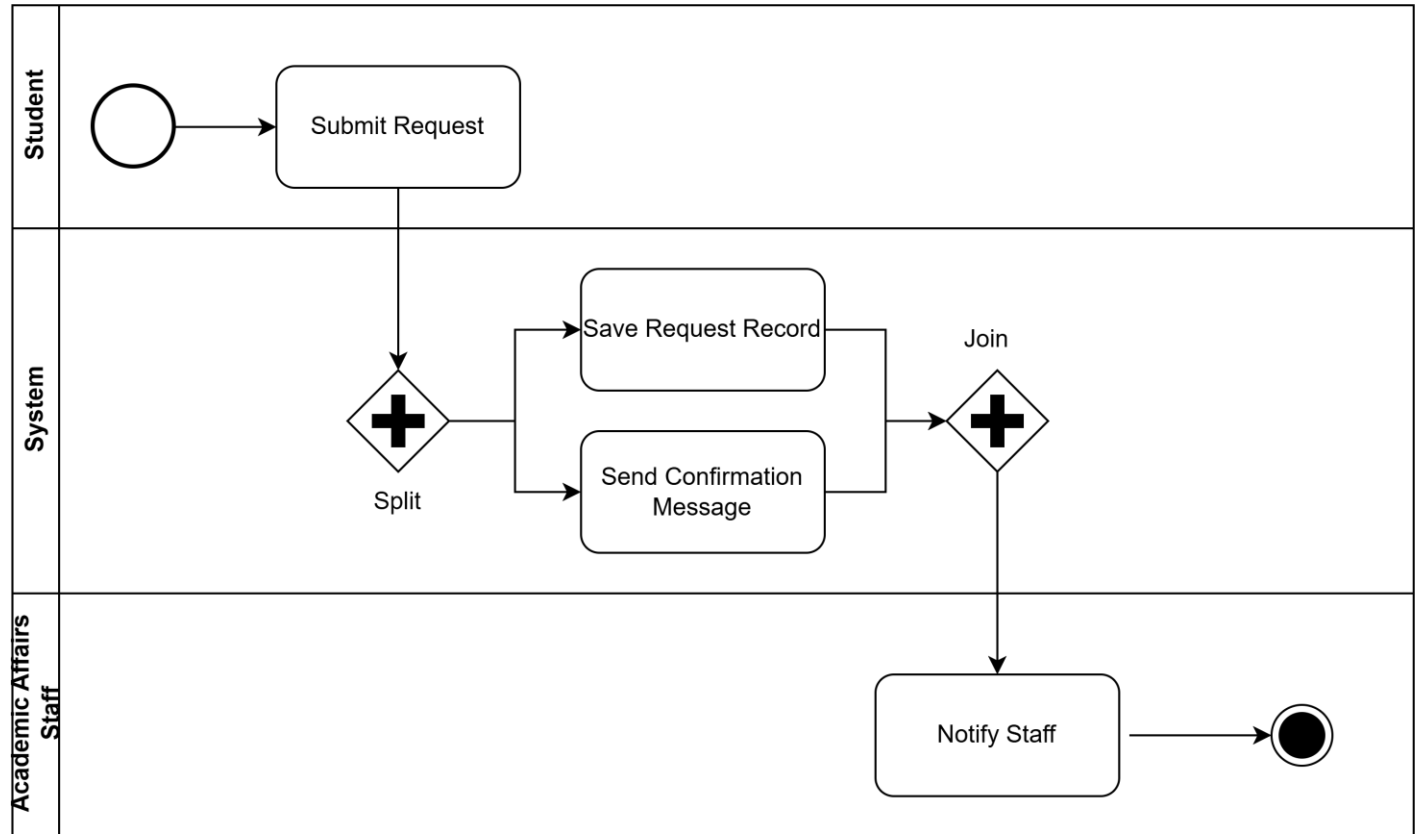
Parallel Gateway

- **A parallel gateway is used when all outgoing paths must be executed.**
- A parallel gateway is used when multiple activities can happen at the same time. The first “+” gateway splits the process into parallel paths, and the second “+” gateway joins the paths before the process continues.
- Do not use AND gateway for Yes/No decisions. Use XOR gateway for alternatives.

Parallel Gateway Example

After a student submits a request, the system may:

- Save Request Record
- Send Confirmation Message



Swimlanes

Swimlanes clarify responsibility.

- **Pool** = major participant or organization
- **Lane** = role, department, or system inside a participant

Example:

- Pool: University
- Lanes:
 - Student
 - Academic Affairs Staff
 - System

Sequence Flow vs Message Flow

Sequence flow may cross lanes inside the same pool. Message flow is used between different pools or independent participants.

Sequence Flow

- Solid arrow
- Shows order of activities
- Used inside the same pool

Message Flow

- Dashed arrow
- Shows communication between different pools
- Used between participants

Important rule:

Do not use message flow between tasks inside the same lane.

Artifacts

Artifacts add information but do not control the process.

Common artifacts:

- **Data Object:** form, document, receipt
- **Annotation:** note or explanation
- **Group:** visually groups related steps

Example:

- Data object: “Uploaded documents”
- Annotation: “Staff must respond within 3 working days”

How to Read a BPMN Diagram

Read BPMN from:

1. Start event
2. Follow sequence flow
3. Check who owns each task by lane
4. Interpret gateway questions
5. Follow Yes/No or parallel paths
6. End at one or more end events

How to Create a BPMN Diagram

Recommended steps:

1. Define process scope
2. Identify participants
3. List main activities
4. Identify decisions
5. Add start and end events
6. Add sequence/message flows
7. Review logic and readability

Common BPMN Mistakes

Avoid:

- Missing start or end event
- Gateway without a question
- Task names without verbs
- Too many crossing arrows
- Using BPMN as DFD
- Drawing system architecture inside BPMN

Summary

BPMN is used to **model business processes**.

Students should **remember**:

- Event = something happens
- Activity = work is done
- Gateway = decision or branching
- Flow = order or communication
- Swimlane = who is responsible

BPMN Practice

Scenario: Student Academic Document Request

A student uses an online portal to request an academic document, such as a transcript or enrollment certificate. The student logs in, fills in the request form, uploads required documents, and submits the request. The system records the request and notifies Academic Affairs staff. Staff review the request. If the information or uploaded documents are incomplete, staff request revision. The student revises the request information, uploads or re-uploads required documents and submits the request again. If complete, staff process the request, update the status, and the system notifies the student.

Practice Requirements

Draw a BPMN diagram with:

- At least 3 lanes:
 - Student
 - System
 - Academic Affairs Staff
- Start event and end event
- At least 5 tasks
- 1 XOR gateway
- Clear sequence flow